Generalization near a bifurcation point

In this work, we attempt to theoretically arrive at the ability of an abstract generative deep neural network to generalize near critical points like divergences, bifurcations, etc when trained on data away from criticality.

1 Linear Toy Model

1.1 Data-generating process

Let $r \in \mathbb{R}$ be a one-dimensional control parameter. We model a bifurcation at $r = r_c$ by introducing a conditional Gaussian distribution whose covariance becomes singular in one direction as $r \to r_c$.

Conditional distribution. For each r, samples $\mathbf{x} \in \mathbb{R}^d$ are drawn from

$$\mathbf{x} \mid r \;\sim\; \mathcal{N}(\mathbf{0},\, \Sigma(r))\,. \tag{1}$$

Critical covariance. Let $\Sigma(r)$ be the covariance of the distribution given in (**??**).
Since the covariance has a symmetric positive semi-definite structure it has an eigen decomposition as follows,

$$\Sigma(r) = Q(r)\Lambda(r)Q(r)^T \tag{2}$$

where

$$\Lambda(r) = \mathrm{diag}(\lambda_1(r), \cdots, \lambda_d(r))$$

Now suppose that as $r \to r_c$, fluctuations in one direction become increasingly dominant. Mathematically,

$$\lambda_1(r) \sim |r - r_c|^{-\nu},$$

while the other eigenvalues remain bounded:

$$\lambda_k(r) = O(1), \qquad k = 2, \cdots, d.$$

Then

$$\frac{\lambda_k(r)}{\lambda_1(r)} \to \infty$$

One direction exhibits parametrically faster growth of variance than the others. [H] Without loss of generality, choose coordinates so that the critical direction is the first canonical basis vector $\mathbf{e}_1 = (1, 0, \ldots, 0)^\top$. Define

$$\Sigma(r) \;=\; I_d + \frac{1}{|r - r_c|^\nu}\,\mathbf{e}_1\mathbf{e}_1^\top, \tag{3}$$

where $\nu > 0$ is the critical exponent. The eigenvalues of $\Sigma(r)$ are

$$\sigma_1(r) \;=\; 1 + |r - r_c|^{-\nu}, \qquad \sigma_k(r) = 1 \quad (k = 2, \ldots, d).$$

As $r \to r_c$, the variance along $\mathbf{e}_1$ diverges as $|r - r_c|^{-\nu}$, mimicking the diverging fluctuations near a critical point. ~~The exponent ν is universal: for the period-doubling cascade in the logistic map at the Feigenbaum point, $\nu = \log 2/\log \delta_F \approx 0.4498$, with $\delta_F \approx 4.6692$ the Feigenbaum constant.~~

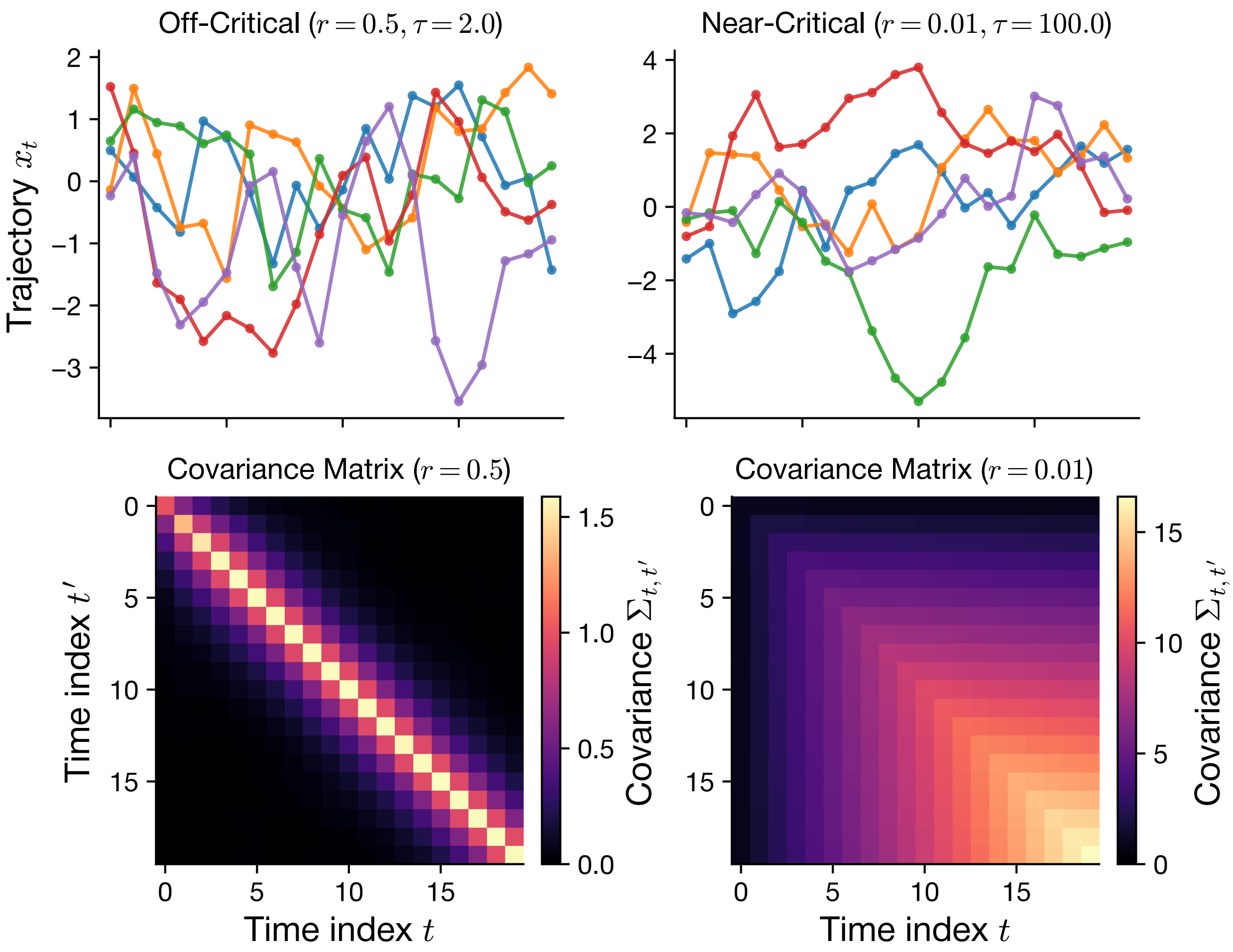


Figure 1: add caption

1.2 True score function

The score of a Gaussian $\mathcal{N}(\mathbf{0}, \Sigma)$ is computed by direct differentiation of its log-density. The density is

$$p(\mathbf{x}) \;=\; \frac{1}{(2\pi)^{d/2}\det(\Sigma)^{1/2}} \exp\!\Big(-\tfrac{1}{2}\mathbf{x}^\top \Sigma^{-1}\mathbf{x}\Big),$$

so

$$\log p(\mathbf{x}) = -\tfrac{1}{2}\mathbf{x}^\top \Sigma^{-1}\mathbf{x} + \text{const}, \qquad \nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\Sigma^{-1}\mathbf{x}.$$

For our problem,

$$s^*(\mathbf{x}, r) \;\equiv\; \nabla_{\mathbf{x}} \log p(\mathbf{x} \mid r) \;=\; -\Sigma(r)^{-1}\mathbf{x}. \tag{4}$$

This inverse can be calculated and we get

$$\Sigma(r)^{-1} \;=\; I_d \,+\, \left(\frac{1}{\sigma_1(r)} - 1\right)\mathbf{e}_1\mathbf{e}_1^\top. \tag{5}$$

Critical direction: $1/\sigma_1(r) = |r - r_c|^\nu/(1 + |r - r_c|^\nu) \to 0$ as $r \to r_c$, so the score component along $\mathbf{e}_1$ vanishes at the bifurcation. **Non-critical directions:** the score is simply $-x_k$ for $k \geq 2$.

2 Student model and loss function

We adopt a linear-in-$\mathbf{x}$ student with general r-dependence:

$$s_{\boldsymbol{\theta}}(\mathbf{x}, r) \;=\; -A(r)\mathbf{x}, \qquad A(r) \in \mathbb{R}^{d\times d}. \tag{6}$$

We parameterize $A(r)$ as a degree-$(K-1)$ polynomial in r in some basis $\{\phi_k(r)\}_{k=0}^{K-1}$ (monomials, Chebyshev, Legendre — the choice does not affect the result). For concreteness:

$$A(r) \;=\; \sum_{k=0}^{K-1} A_k\, \phi_k(r), \qquad \phi_k(r) = r^k.$$

The trainable parameters are the K matrices $\{A_k\}$, totaling Kd^2 scalar parameters.

The (population) score-matching loss is

$$\mathcal{L}(\boldsymbol{\theta}) \;=\; \mathbb{E}_{r\sim\rho_\Delta}\left[\mathbb{E}_{\mathbf{x}|r}\left[\left\|s_{\boldsymbol{\theta}}(\mathbf{x}, r) - s^*(\mathbf{x}, r)\right\|^2\right]\right], \tag{7}$$

where the training distribution over r has the gap excised:

$$\rho_\Delta(r) \;=\; \frac{1}{Z_\Delta}\,\mathbf{1}[\,|r - r_c| > \Delta/2\,]\cdot \rho_0(r), \tag{8}$$

with ρ_0 a base distribution (we take uniform on $[r_c - L, r_c + L]$) and Z_Δ the normalization.

Let $B(r) \equiv A(r) - \Sigma(r)^{-1}$. Then

$$\left\|s_{\boldsymbol{\theta}}(\mathbf{x}, r) - s^*(\mathbf{x}, r)\right\|^2 \;=\; \left\|B(r)\mathbf{x}\right\|^2 \;=\; \mathbf{x}^\top B(r)^\top B(r)\mathbf{x}.$$

For any deterministic symmetric matrix M, the standard identity for a centered Gaussian $\mathbf{x} \sim \mathcal{N}(0, \Sigma)$ is $\mathbb{E}[\mathbf{x}^\top M\mathbf{x}] = \mathrm{Tr}(M\Sigma)$. Hence

$$\mathbb{E}_{\mathbf{x}|r}\left[\|s_{\boldsymbol{\theta}} - s^*\|^2\right] \;=\; \mathrm{Tr}\big(B(r)^\top B(r)\Sigma(r)\big) \;=\; \mathrm{Tr}\big(B(r)\Sigma(r)B(r)^\top\big).$$

So the loss becomes

$$\mathcal{L}(\boldsymbol{\theta}) \;=\; \int \rho_\Delta(r)\ \mathrm{Tr}\Big[\big(A(r) - \Sigma(r)^{-1}\big)\,\Sigma(r)\,\big(A(r) - \Sigma(r)^{-1}\big)^\top\Big]\,\mathrm{d}r. \tag{9}$$

2.1 Decoupling the dimensions

We now exploit the diagonal structure of $\Sigma(r)$. In the canonical basis, $\Sigma(r) = \mathrm{diag}(\sigma_1(r), 1, 1, \ldots, 1)$. The cost (**??**) can be expanded in matrix entries. Off-diagonal entries of $A(r)$ contribute only positive terms to the trace, while $\Sigma(r)^{-1}$ has no off-diagonal part. Hence setting off-diagonal entries of $A(r)$ to zero strictly reduces the cost without affecting the diagonal fit. Writing $A(r) = \mathrm{diag}(a_1(r), a_2(r), \ldots, a_d(r))$ with each $a_k(r)$ a degree-$(K-1)$ polynomial, the trace splits as

$$\mathrm{Tr}[\cdots] \;=\; \sum_{k=1}^{d}\big(a_k(r) - 1/\sigma_k(r)\big)^2\sigma_k(r).$$

Non-critical directions ($k \geq 2$). Here $\sigma_k(r) = 1$ and $1/\sigma_k(r) = 1$, so the k-th term is $(a_k(r) - 1)^2$. The polynomial fit $a_k(r) \equiv 1$ (achievable since the constant function belongs to the polynomial basis) gives zero loss. These directions are trivially learned.

Critical direction ($k = 1$). Drop the subscript and call this $a(r) \equiv a_1(r)$. The relevant loss is

$$\mathcal{L}_1(a) \;=\; \int \rho_\Delta(r)\,\big(a(r) - 1/\sigma_1(r)\big)^2 \sigma_1(r)\,\mathrm{d}r. \tag{10}$$

Define the *target function* and *weight function*:

$$g(r) \;\equiv\; \frac{1}{\sigma_1(r)} \;=\; \frac{|r - r_c|^\nu}{1 + |r - r_c|^\nu}, \qquad w(r) \;\equiv\; \rho_\Delta(r)\sigma_1(r). \tag{11}$$

Then

$$\mathcal{L}_1(a) \;=\; \int w(r)\,\big(a(r) - g(r)\big)^2\,dr. \tag{12}$$

This is a *weighted least-squares polynomial fit* of g on the support of ρ_Δ (i.e., on $|r - r_c| > \Delta/2$). Note that

- For $|r - r_c| \ll 1$: $g(r) \approx |r - r_c|^\nu$ — the cusp.
- For $|r - r_c| \to \infty$: $g(r) \to 1$ — saturates.
- At $r = r_c$: $g(r_c) = 0$.

3 Minimising loss

We henceforth set $r_c = 0$ by relabeling. Expand $a(r) = \sum_{k=0}^{K-1} \theta_k r^k$, with $\boldsymbol{\theta} = (\theta_0, \dots, \theta_{K-1})^\top$ and $\boldsymbol{\phi}(r) = (1, r, r^2, \dots, r^{K-1})^\top$, so

$$a(r) \;=\; \boldsymbol{\theta}^\top \boldsymbol{\phi}(r).$$

The loss becomes a quadratic form in $\boldsymbol{\theta}$:

$$\begin{aligned} \mathcal{L}_1 \;&=\; \int w(r)\,\big(\boldsymbol{\theta}^\top \boldsymbol{\phi}(r) - g(r)\big)^2\,dr \\ &=\; \boldsymbol{\theta}^\top \Big[\int w(r)\boldsymbol{\phi}(r)\boldsymbol{\phi}(r)^\top dr\Big]\boldsymbol{\theta} \;-\; 2\boldsymbol{\theta}^\top \Big[\int w(r)g(r)\boldsymbol{\phi}(r)dr\Big] \;+\; \int w(r)g(r)^2 dr. \end{aligned} \tag{13}$$

Define

$$M_{kl} \;\equiv\; \int w(r)\,r^k r^l\,dr \;=\; \int \rho_\Delta(r)\sigma_1(r)\,r^{k+l}\,dr, \tag{14}$$

$$b_k \;\equiv\; \int w(r)g(r)\,r^k\,dr. \tag{15}$$

Crucially, $\sigma_1(r)g(r) = 1$, so the bias vector simplifies:

$$b_k \;=\; \int \rho_\Delta(r)\,r^k\,dr. \tag{16}$$

That is, b_k is the k-th moment of the *unweighted* training distribution ρ_Δ.

Setting $\partial\mathcal{L}_1/\partial\boldsymbol{\theta} = 0$ gives the normal equations $M\boldsymbol{\theta} = \mathbf{b}$, with solution

$$\hat{\boldsymbol{\theta}} \;=\; M^{-1}\mathbf{b}. \tag{17}$$

4 Scaling analysis

Place $r_c = 0$. Define a matching scale r_* of order unity that separates inner and outer regions:

- **Inner region:** $\Delta/2 < |r| < r_*$, with $|r|^\nu \ll 1$, so $\sigma_1(r) \approx |r|^{-\nu}$ and $g(r) \approx |r|^\nu$.
- **Outer region:** $r_* < |r| < L$, with $|r|^\nu \gtrsim 1$, so $\sigma_1(r) \to 1$ and $g(r) \to 1$.

4.1 Inner-region contributions

Assume $k+l$ even (only nonvanishing cases). The inner contribution to M_{kl} is

$$\begin{aligned} M_{kl}^{\text{in}}(\Delta) &= \int_{\Delta/2<|r|<r_*} \rho_0(r)\sigma_1(r)\, r^{k+l}\, dr \\ &\approx 2\rho_0 \int_{\Delta/2}^{r_*} r^{-\nu}\, r^{k+l}\, dr \\ &= \frac{2\rho_0}{k+l-\nu+1} \Big[r_*^{k+l-\nu+1} - (\Delta/2)^{k+l-\nu+1} \Big]. \end{aligned} \tag{18}$$

For $\nu < 1$ (Feigenbaum case) and $k+l \geq 0$ even, the exponent $k+l-\nu+1 > 0$, so the integral is finite as $\Delta \to 0$ and the Δ-dependence enters as

$$M_{kl}^{\text{in}}(\Delta) = M_{kl}^{\text{in}}(0) - \frac{2\rho_0}{k+l-\nu+1} (\Delta/2)^{k+l-\nu+1}. \tag{19}$$

Similarly,

$$\begin{aligned} b_k^{\text{in}}(\Delta) &= \int_{\Delta/2<|r|<r_*} \rho_0(r)\, r^k\, dr \\ &\approx \frac{2\rho_0}{k+1} \Big[r_*^{k+1} - (\Delta/2)^{k+1} \Big]. \end{aligned} \tag{20}$$

4.2 Outer-region contributions

For the outer region, $\sigma_1(r) \to 1$, $g(r) \to 1$, $w(r) \to \rho_0$:

$$M_{kl}^{\text{out}} \approx \rho_0 \int_{r_*<|r|<L} r^{k+l}\, dr, \qquad b_k^{\text{out}} \approx \rho_0 \int_{r_*<|r|<L} r^k\, dr. \tag{21}$$

Both are Δ-independent at this level of approximation.

4.3 Expansion of $\hat{\boldsymbol{\theta}}(\Delta)$

Combining,

$$M(\Delta) = M_0 + \Delta^{1-\nu} M_1 + O(\Delta^{3-\nu}), \tag{22}$$

$$\mathbf{b}(\Delta) = \mathbf{b}_0 + \Delta\, \mathbf{b}_1 + O(\Delta^3), \tag{23}$$

where $M_0, M_1, \mathbf{b}_0, \mathbf{b}_1$ are Δ-independent. The corrections in M scale as $\Delta^{1-\nu}$ because the leading inner integrand $\sim r^{-\nu}$ near the gap edge integrates to give a power of the gap that is one greater (boundary contribution). The corrections in $\mathbf{b}$ scale as Δ because the integrand is regular ($\sim r^0$).

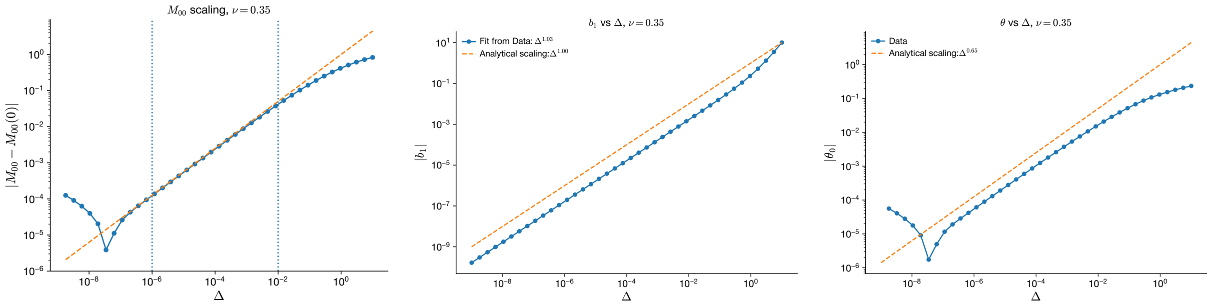
Hence

$$\begin{aligned} \hat{\boldsymbol{\theta}}(\Delta) &= M(\Delta)^{-1}\mathbf{b}(\Delta) \\ &= \big(M_0 + \Delta^{1-\nu} M_1 + \cdots\big)^{-1} \big(\mathbf{b}_0 + \Delta\, \mathbf{b}_1 + \cdots\big) \\ &= M_0^{-1}\mathbf{b}_0 + \Delta^{1-\nu} \Big[- M_0^{-1} M_1 M_0^{-1} \mathbf{b}_0 \Big] + O(\Delta). \end{aligned} \tag{24}$$

So

$$\hat{\theta}_0(\Delta) = \hat{\theta}_0^{(0)} + c_1\, \Delta^{1-\nu} + O(\Delta), \tag{25}$$

where $\hat{\theta}_0^{(0)} = \hat{\boldsymbol{\theta}}_0^{(0)}$ is the limiting value at $\Delta \to 0$, with $\hat{\boldsymbol{\theta}}^{(0)} = M_0^{-1}\mathbf{b}_0$.



(a) (b) (c)

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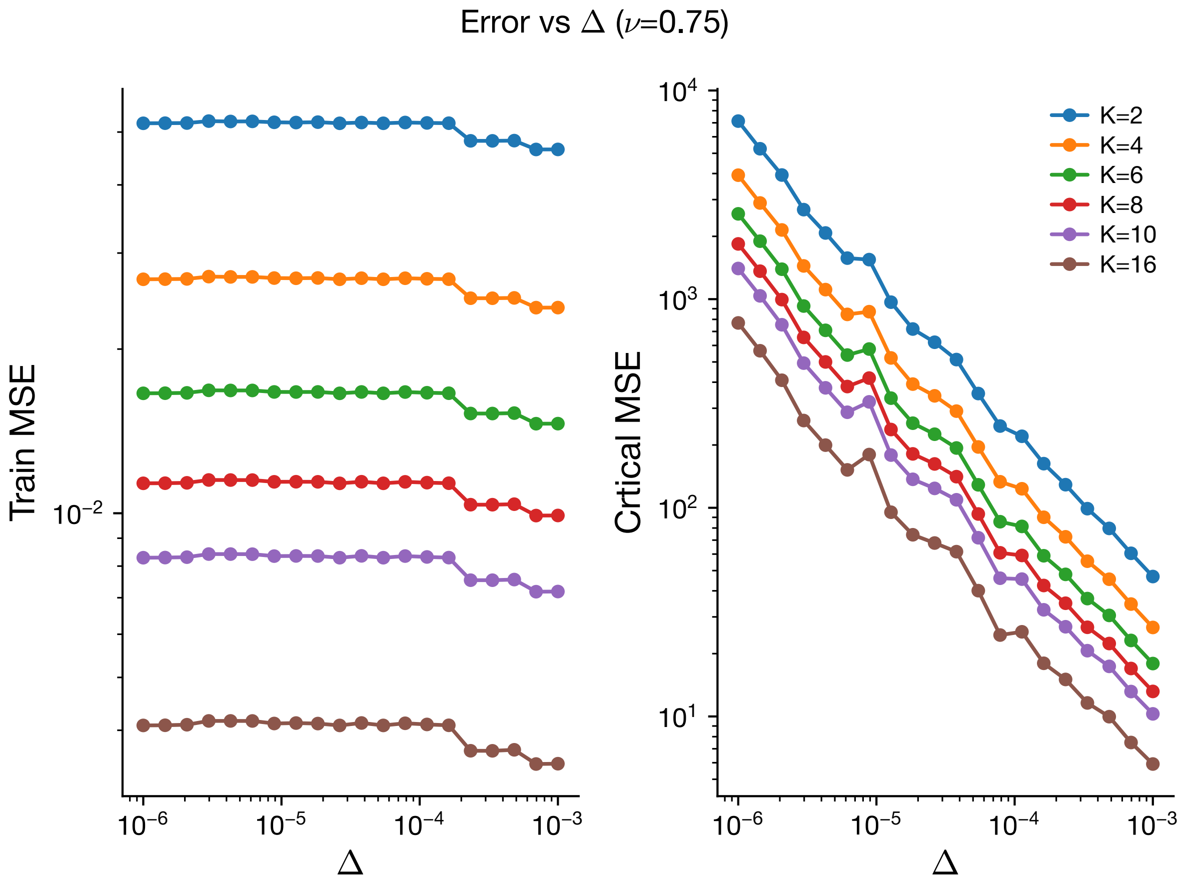

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5 Finite sampling effects for a deterministic target

Now we move on to analyzing the scaling of θ that solves the following problem:

$$\mathcal{L} = \frac{1}{2}\sum_{\mu=1}^{P}\left[w(r_\mu)\left(\hat{\theta}^T\phi(r_\mu) - y_\mu\right)^2\right] + \frac{\lambda}{2}||\hat{\theta}||^2 \tag{26}$$

The loss function in (**??**) has been modified to accommodate finite number of data points P.

Differentiating the loss function with respect to $\hat{\theta}$ and setting to zero, we arrive at

$$\sum_{\mu=1}^{P} w(r_\mu)\phi(r_\mu)\phi(r_\mu)^T\hat{\theta} - \sum_{\mu=1}^{P} w(r_\mu)y_\mu\phi(r_\mu) + \lambda\hat{\theta} = 0$$

$$\implies \left(\frac{1}{P}\sum_{\mu=1}^{P} w(r_\mu)\phi(r_\mu)\phi(r_\mu)^T + \frac{\lambda}{P}I\right)\hat{\theta} = \frac{1}{P}\sum_{\mu=1}^{P} w(r_\mu)y_\mu\phi(r_\mu)$$

This motivates the definitions

$$\hat{M} = \frac{1}{P}\sum_{\mu=1}^{P} w(r_\mu)\phi(r_\mu)\phi(r_\mu)^T \tag{27}$$

and

$$\hat{b} = \frac{1}{P}\sum_{\mu=1}^{P} w(r_\mu)y_\mu\phi(r_\mu) \tag{28}$$

So the empirical estimator can be compactly written as

$$\hat{\theta}(\Delta, K, P) = \big(\hat{M} + \frac{\lambda}{P}I\big)^{-1}\hat{b} \tag{29}$$

$$\hat{b} \in \mathbb{R}^{K\times 1}; \qquad \hat{M} \in \mathbb{R}^{K\times K}; \qquad \hat{\theta} \in \mathbb{R}^{K\times 1} \tag{30}$$

Assume

$$y_\mu = g(r_\mu) \tag{31}$$

Therefore

$$\hat{b} = \frac{1}{P}\sum_{\mu=1}^{P} w(r_\mu)g(r_\mu)\phi(r_\mu) \tag{32}$$

and

$$\hat{M} = \frac{1}{P}\sum_{\mu=1}^{P} w(r_\mu)\phi(r_\mu)\phi(r_\mu)^T \tag{33}$$

Then the estimator is

$$\hat{\theta} = \big(\hat{M} + \frac{\lambda}{P}I\big)^{-1}\hat{b} \tag{34}$$

$$\implies \hat{\theta}(\Delta, K, P) = \left(\frac{1}{P}\sum_{\mu=1}^{P} w(r_\mu)\phi_\mu\phi_\mu^T + \frac{\lambda}{P}I\right)^{-1}\left(\frac{1}{P}\sum_{\mu=1}^{P} w(r_\mu)y_\mu\phi_\mu\right) \tag{35}$$

Define

$$M(\Delta, K) = \mathbb{E}_{r\sim\rho_\Delta}[\hat{M}] = \mathbb{E}_{r\sim\rho_\Delta}[w(r)\phi(r)\phi(r)^T] \tag{36}$$

$$b(\Delta, K) = \mathbb{E}_{r\sim\rho_\Delta}[\hat{b}] = \mathbb{E}_{r\sim\rho_\Delta}[w(r)g(r)\phi(r)] \tag{37}$$

As $P \to \infty$

$$\hat{\theta}(\Delta, K, P) \to \theta_\infty(\Delta, K) = M^{-1}b \tag{38}$$

which corresponds to the continuum solution obtained in the infinite-sample limit(see eqn (**??**)).

5.1 Perturbation around infinite sample limit

We now introduce the finite data effect as a perturbation to the infinite limit case.

$$\hat{M} = M(\Delta, K) + \delta M \tag{39}$$

and

$$\hat{b} = b(\Delta, K) + \delta b \tag{40}$$

where

$$\delta M = \frac{1}{P}\sum_{\mu=1}^{P}[w(r_\mu)\phi_\mu\phi_\mu^T - M]; \qquad \delta b = \frac{1}{P}\sum_{\mu=1}^{P}[w(r_\mu)g(r_\mu)\phi_\mu - b]$$

The mean and variance of δM and δb are as follows:

$$\mathbb{E}[\delta M] = \mathbb{E}[\delta b] = 0 \tag{41}$$

and

$$\mathbb{E}[\delta M \delta M^T] = \mathbb{E}[\delta b \delta b^T] = O(P^{-1}) \tag{42}$$

$$\implies \delta M \sim O(P^{-1/2}), \qquad \delta b \sim O(P^{-1/2}) \tag{43}$$

Further we define,

$$A = M + \frac{\lambda}{P}I \tag{44}$$

Then,

$$\hat{\theta}(\Delta, K, P) = (A + \delta M)^{-1}(b + \delta b) = \big(A^{-1} - A^{-1}\delta M A^{-1} + O(\delta M^2)\big)(b + \delta b) \tag{45}$$

$$\implies \hat{\theta} = A^{-1}b + A^{-1}\delta b - A^{-1}\delta M A^{-1} b + O(P^{-1}) \tag{46}$$

$$\implies \hat{\theta} = \theta + A^{-1}\delta b - A^{-1}\delta M \theta + O(P^{-1}) \tag{47}$$

$$\implies \hat{\theta}(\Delta, K, P) = \theta(\Delta, K) + \frac{1}{\sqrt{P}}\eta(\Delta, K) + O(P^{-1}) \tag{48}$$

where

$$\eta(\Delta, K) = \sqrt{P}(A^{-1}\delta b - A^{-1}\delta M\theta) \tag{49}$$

The first and second moments of η can be found to be,

$$\mathbb{E}[\eta] = \sqrt{P}A^{-1} \qquad \mathbb{E}[\delta b] - A^{-1} \qquad \mathbb{E}[\delta M\theta] = 0 \tag{50}$$

$$\mathbb{E}[\eta\eta^T] = \mathbb{E}[P(A^{-1}(\delta b - \delta M\theta))(A^{-1}(\delta b - \delta M\theta))^T] \tag{51}$$

$$= PA^{-1}\mathbb{E}[\delta b\delta b^T - \delta b(\delta M\theta)^T - (\delta M\theta)\delta b^T + (\delta M\theta)(\delta M\theta)^T(A^{-1})^T] \tag{52}$$

The expectation evaluates to an $O(P^{-1})$ quantity. Therefore

$$\mathbb{E}[\eta\eta^T] = O(1)$$

Now we can arrive at the first and second moments of $\hat{\theta}(\Delta, K, P)$

$$\mathbb{E}[\hat{\theta}] = \theta(\Delta, K) + O(P^{-1})$$

6 Finite Sampling with noisy target

$$r_\mu \overset{i.i.d}{\sim} \rho_\Delta,$$

$$y = g(r_\mu) + \xi_\mu; \quad \xi \sim \mathcal{N}(0, \tau^2),$$

and

$$\phi(r) = (1, r, r^2, ..., r^{K-1})^T$$